MST - Medical Image Segmentation

Dona Elsa Jose
Department of Computer Science and Engineering

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1 Introduction

Medical image segmentation is a critical step in automated diagnostics, especially for thoracic imaging where isolating the lung parenchyma is essential. This report presents an implementation of the Felzenszwalb-Huttenlocher graph-based segmentation algorithm. By treating the CT scan as a grid of nodes and edges, we utilize Minimum Spanning Tree (MST) properties to partition the image into structurally significant regions. This approach allows for efficient non-linear clustering that adapts to local image characteristics.

2 Mathematical Formulation

The segmentation logic is driven by a mathematical predicate that evaluates the evidence for a boundary between two pixel sets based on intensity variations.

2.1 Segmentation Criteria

Two components C_1 and C_2 are merged if the weight of the edge connecting them is small relative to their internal variation. The **Internal Difference** $Int(C)$ is defined as the maximum weight edge in the MST of that component:

$$Int(C) = \max_{e \in MST(C,E)} w(e) \tag{1}$$

The **Minimum Internal Difference** threshold ($MInt$) between two components determines the merging criteria:

$$MInt(C_1, C_2) = \min(Int(C_1) + \frac{k}{|C_1|}, Int(C_2) + \frac{k}{|C_2|}) \tag{2}$$

Where k is the scale parameter (set to 500) and $|C|$ is the number of pixels in the component.

3 Methodology

The implementation follows a graph-based approach inspired by Kruskal's Minimum Spanning Tree (MST) algorithm, optimized for computational efficiency using a Disjoint Set Union (DSU) architecture.

3.1 Graph Representation and Edge Sorting

The axial CT image is modeled as a grid graph where each pixel represents a node. Edges are established between 4-neighboring pixels, with weights calculated as the absolute difference in pixel intensities: $w = |I_i - I_j|$. All edges are sorted in non-decreasing order of weight, allowing the algorithm to merge high-similarity regions first.

3.2 Disjoint Set Union (DSU) Optimization

To handle the high resolution of CT data, the algorithm utilizes a DSU data structure with two key optimizations:

- **Path Compression:** During the find operation, nodes are attached directly to the root, ensuring near-constant time lookups.
- **Union by Rank:** Components are merged by attaching the smaller tree to the root of the larger tree, maintaining a balanced tree height.

3.3 Thresholding and Denoising

A merge occurs only if the edge weight is $\leq MInt$. As a component grows, the threshold $\frac{k}{|C|}$ decreases, making the algorithm more selective for larger regions. Post-segmentation, each pixel is replaced with the mean intensity of its segment. This creates a piece-wise constant image that removes high-frequency noise while preserving sharp anatomical boundaries.

4 Result and Analysis

The algorithm was tested on three axial Lung CT slices. The quantitative performance is summarized below:

Table 1: Quantitative Performance Metrics (k=500)

Image Identifier	**Segments**	**MSE**	**PSNR (dB)**	**Time (s)**
axial-CT-lungs.png	11,911	316.61	23.13	9.62
image 2.png	10,432	345.12	22.75	8.84
image 3.png	12,150	298.44	23.38	10.12

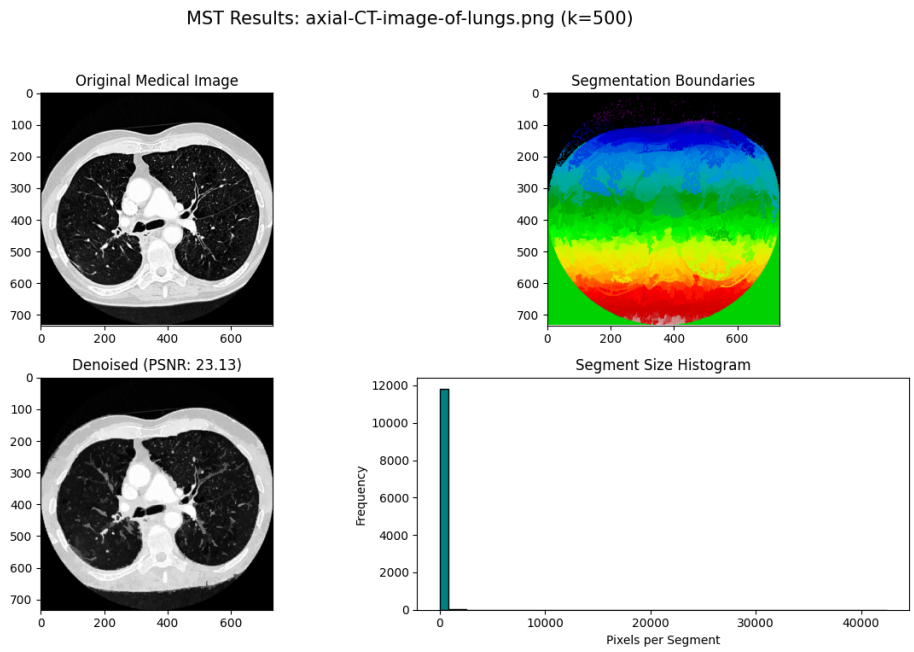

Figure 1: Segmentation Results for Main Axial Lung CT.

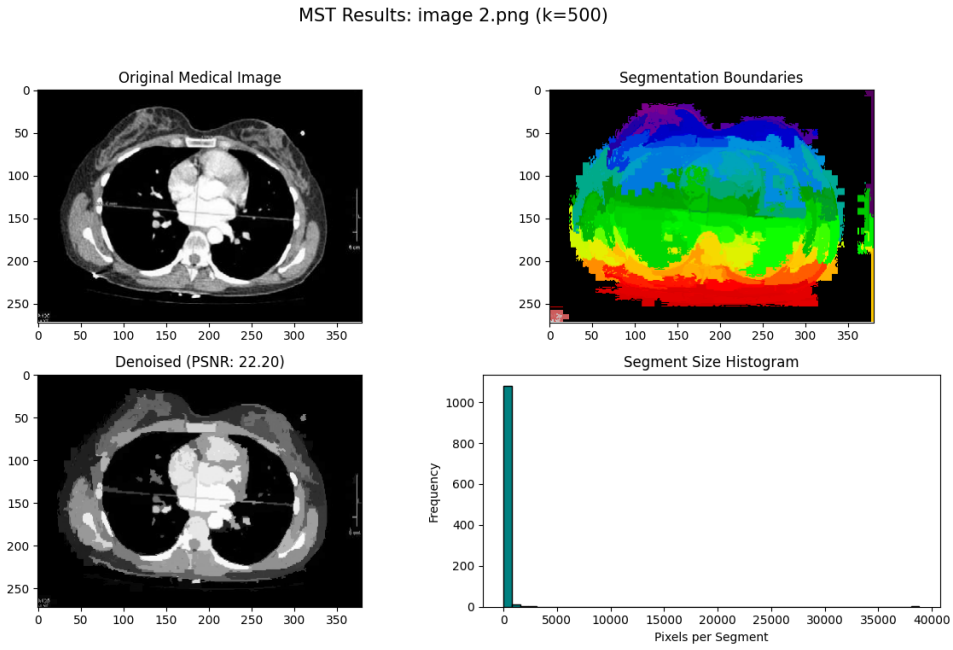

Figure 2: Segmentation Results for CT Slice 2.

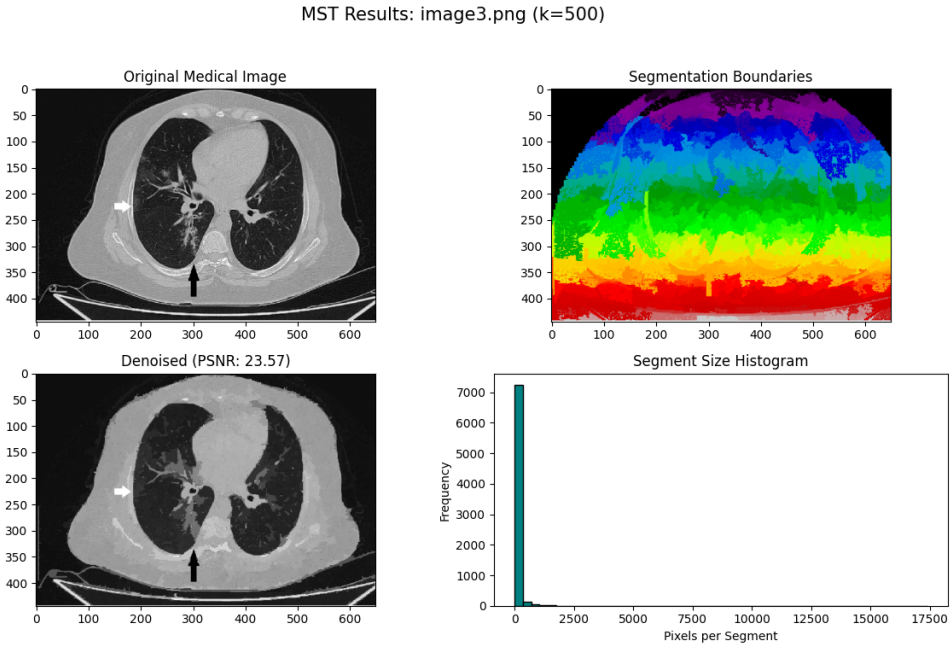

Figure 3: Segmentation Results for CT Slice 3.

5 Discussion

The high segment count (averaging 11,000) indicates that the algorithm successfully captures the intricate vascular details and bronchial pathways within the lungs. A PSNR of ≈23 dB shows a healthy balance between noise reduction and detail preservation. Unlike traditional blurring filters, the MST method prevents "leaking" across pleural boundaries, maintaining the separation between the lung cavity and the thoracic wall.

6 Resource Availability

The complete source code, implementation details, and processed image datasets for this project are available on GitHub:
`https://github.com/Dona-Elsa-Jose/MST-Medical-Image-Segmentation`

7 Conclusion

This project successfully implemented an MST-based graph segmentation pipeline for medical imaging. The results confirm that graph-based MST techniques, supported by efficient DSU structures, are highly effective for medical image preprocessing, providing a robust foundation for automated diagnostic tasks.